

LeWAM: A Latent World-Action Model

When does latent world-model planning help for manipulation?

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Abstract

We study *when* a latent world model can plan for contact-rich manipulation. We introduce **LeWAM**, a latent world-action model that augments a JEPA latent world model with a jointly trained, history-conditioned action pathway, yielding a single model that both predicts latent dynamics and proposes actions. In contrast to the dominant world-action models—whether video-generative (DreamZero) or latent (Being-H0.7)—which learn to *avoid* test-time search and amortize planning into one forward pass, LeWAM deliberately retains short-horizon planning and asks *when it helps*. Our central result is an analysis of when latent world-model planning helps: short-horizon CEM against a goal-image latent cost succeeds on goal-reaching tasks where the goal encodes a large, visible, data-supported state change (OGBench-Cube, Push-T, Reacher), but collapses to near-zero on contact-rich robomimic (Lift/Can/Square), where planning fails for two distinct reasons we separate with probes: on Lift the success variable (grasp) is visually subdominant in a single frame, while on Can/Square the latent does see grasp but the task is multi-stage and a flat goal-image cost cannot decompose it. We show that a history / past-action-conditioned behavior-cloning pathway recovers this grasp state and far outperforms online planning on contact. Through a multi-training-seed study we further find that, at convergence, the action-head objective (detached vs. grounded decoder, raw-action vs. embedding-target, anti-collapse regularization) does not move success rate beyond training-run noise—a cautionary negative result. LeWAM’s contribution is therefore the per-family planning analysis and the history-conditioned action pathway, not a state-of-the-art success rate.

1 Introduction

From behavior cloning to world-action models. [TODO: 1 para: VLA/BC map obs→action with no explicit dynamics; the field is moving to world-action models (WAMs) that jointly model world + action, giving a forward model to plan/verify with. Two flavors: video-generative (DreamZero [24]) and latent (the LeWM [15] lineage). LeWAM is the latent WAM.]

The question. Latent world models plan by sampling-based search (CEM/MPC) against a latent goal cost. This works on some tasks and fails on others; *when, and why?* [TODO: frame the per-family question as the paper’s spine.]

Contributions.

- **A per-family analysis of latent WM-planning.** We show CEM planning succeeds on goal-reaching/geometry tasks and collapses to ≈ 0 on contact-rich robomimic, and we identify the mechanism: the goal-image latent cost rewards the task only when the goal encodes a large, visible, data-supported state change (§4.1).
- **A history-conditioned action pathway** that far outperforms online planning on all contact tasks; on Lift it provably recovers grasp state that a single frame exposes only weakly

(probe AUC 0.82 vs 0.99 from history), while on multi-stage Can/Square it supplies the sequential execution a flat planner lacks (§4.2).

- **An honest multi-training-seed account:** at convergence the action-head objective does not change success rate beyond training-run noise; we make no SOTA-SR claim and instead position on mechanism and analysis (§4.3).
- **The LeWAM reframe:** a JEPA world model and its policy head as a single latent world-action model, the lightweight latent counterpart to video-generative WAMs.

2 Related Work

World-action models: video-generative versus latent. A *world-action model* (WAM) jointly models environment dynamics and the action that drives them, $p(\text{future}, \text{action})$, so that a single learned object both predicts what happens next and supplies the control to make it happen. This frames the WAM against two neighbors. Unlike a vision-language-action policy [12, 4], which maps observations directly to actions with no explicit forward model, a WAM carries dynamics and can reason about counterfactual futures. Unlike a pure latent or 3D world model [26, 17], which predicts futures but defers action selection to a separate inverse model or to test-time search, a WAM internalizes the action pathway. The family splits on the medium of the future. *Video-generative* WAMs, exemplified by DreamZero [24], model $p(\text{video}, \text{action})$ with video diffusion, inherit physics priors from internet-scale video, and learn inverse dynamics from their own predicted rollouts so that no test-time optimization is required. *Latent* WAMs, the LeWM [15] lineage we extend, predict in a self-supervised joint-embedding latent rather than in pixels, keeping the model light and the dynamics abstract. Being-H0.7 [14] is a recent large-scale latent WAM, organizing a future-aware latent world-action workspace from 200k hours of egocentric video; like DreamZero it is search-free, discarding its training-only posterior branch and performing no rollout at inference. A common thread thus emerges: the dominant WAMs, whether video-generative (DreamZero) or latent (Being-H0.7), *learn to avoid test-time search*, amortizing planning into a single forward pass. Fast-WAM [25] makes the trend explicit, asking whether world-action models need test-time future imagination at all and answering *no*—uniformly—finding that video co-training rather than test-time rollout drives control. LeWAM is also a latent WAM—a JEPA predictor with a jointly trained action pathway—but it revisits this question for latent models with *explicit* search and answers it *per task family* rather than uniformly: it deliberately retains a short-horizon search loop and asks *when does latent world-model planning actually help, and when should one fall back to a history-conditioned policy?* Where Fast-WAM removes test-time imagination wholesale, LeWAM shows planning still pays off on goal-reaching/geometry tasks and collapses on contact-rich manipulation, and supplies the history-conditioned pathway that carries the latter case. Its contribution is therefore not to be a latent WAM (Being-H0.7 already is one at scale) but to characterize, per task family, where planning over a latent world model pays off and where it collapses—and to supply the history-conditioned pathway that carries the latter case.

JEPA and latent world models for control. A growing line of work learns a world model in a self-supervised joint-embedding (JEPA) latent and plans within it, dispensing with pixel reconstruction. LeWM [15], the latent on which we build, trains a decoder-free predictor for pixel-latent consistency and plans by sampling-based optimization (CEM) against a latent goal-image cost; DINO-WM [26] plans the same way over frozen DINO features with explicitly “no pre-learned inverse models,” and V-JEPA 2-AC [17] extends a video JEPA to a real Franka arm, again with pure

CEM at action time. While these systems convincingly show that a JEPA latent supports planning across navigation, deformables, and continuous control, none carries a learned action prior: the shared pattern is *pretrained features plus sampling-based CEM/MPC with no action model*, which leaves every candidate rollout at the mercy of a dynamics model never trained to model contact. The field supplies its own evidence that this is brittle, V-JEPA 2-AC’s grasp success on a physical arm sitting well below its reaching performance, exactly the regime where rollouts must traverse contact dynamics the latent does not capture. A complementary line improves the latent’s *geometry* for planning rather than its action model: temporal straightening [22] regularizes the JEPA representation so that latent trajectories are locally straight and Euclidean cost approximates geodesic distance, sharpening the goal-image planning landscape and raising goal-reaching success. This is orthogonal to what we find: straightening conditions a geometry in which the task-relevant variable is already present, whereas our robomimic failures are of a different kind that no metric-geometry regularizer addresses—on Lift the success variable (grasp) is *under-exposed* in the single frame, and on Can/Square it is present yet the task is *multi-stage* so a flat goal-image cost cannot decompose it—both recovered instead by our history-conditioned pathway executing the demonstrated sequence. In contrast, LeWAM keeps this JEPA substrate but attaches a jointly trained action pathway and studies, per task family, where the substrate’s planner does and does not work.

Learned policy as a planning prior. A second tradition warm-starts search with a learned policy, the “thinking fast and slow” division of labor introduced by expert iteration [2]. TD-MPC2 [8] and its predecessor [10] learn a Markovian policy $\pi(a | z_t)$ on a decoder-free latent and use it to seed MPPI, and the MuZero family [21] plays the same role for discrete-action MCTS. Recent instances follow suit: policy-guided world-model planning for language-conditioned navigation [5] seeds search from an instantaneous policy, and the JEPA-native Newt [9] pairs a behavior-cloned prior with annealed-constraint planning across a massively multitask, padded-and-masked action space. These priors successfully turn an expensive search into a guided one, but across the cluster the prior is *instantaneous and raw-action*: it conditions on the current latent state and proposes a single next action. The sole precedent that conditions on a past action, MBOP [3] with $a_t = f_b(s_t, a_{t-1})$, is feed-forward in one previous action, non-JEPA, and raw-action; DINO-WM and V-JEPA 2-AC condition their *dynamics* on action history yet carry no prior at all. In contrast, LeWAM’s action pathway is *autoregressive over past actions* $a_{<t}$ on a JEPA latent rather than Markovian, which is what lets it recover the contact state (“am I holding the object”) that a single-frame latent cannot expose. Newt and LeWAM both predict actions on JEPA latents; LeWAM is distinguished by the history and past-action conditioning and by the planning analysis that motivates it.

Amortized and search-free latent action. A third position replaces search outright by directly predicting actions in latent space. The most relevant instance is GC-IDM [18], a goal-conditioned inverse-dynamics model trained on the *same LeWM latent*, which maps (current latent, goal latent, remaining horizon) to the next action and matches or exceeds CEM across navigation, manipulation, and control at roughly two orders of magnitude lower per-decision cost; Diffuser [11] and Decision Diffuser [1] likewise generate trajectories in one shot from return or skill guidance, and DPPO [20] makes the policy itself the planner. These methods compellingly show that amortization can match search at a fraction of the cost, but in making amortization the destination they forfeit any test-time correction when a single forward pass is wrong. In contrast, LeWAM treats amortization as a *warm start to be refined*, with the action pathway proposing and short-horizon search correcting it where a single inverse-dynamics pass is brittle. GC-IDM, sharing our base, is the must-beat

amortization baseline against which the warm-start-then-refine wedge has to earn its cost.

Hierarchical, diffusion-execution, and manipulation planners. On contact-rich manipulation specifically, the dominant recipe is to let a world model plan subgoals and a learned policy execute them. WorldDP [7] diagnoses our setting directly, noting that visual world-model MPC is “limited to single-stage tasks such as reaching or grasping” and struggles with multi-stage ones, and its fix is hierarchical: a high-level world model that optimizes feasible subgoals at runtime with a low-level diffusion policy reaching each, over object-centric slot representations. LDP [23] is the closest published latent planner on our benchmark, reporting robomimic image Lift/Can/Square with a diffusion forecaster and diffusion inverse dynamics rather than a JEPA latent plus CEM, while FF-JEPA [16], Diffusion Policy [6], and VQ-BeT [13] round out the execution-side toolkit of expressive, multimodal action models. These planners convincingly solve multi-stage manipulation, but each assumes the failure of flat latent planning rather than mapping it. In contrast, LeWAM proposes no new hierarchy; its contribution on this axis is a per-family analysis of *when flat latent planning fails*. Single-frame goal-image CEM succeeds when the goal encodes a large, visible, data-supported state change (e.g. an OGBench cube carried to a new pose) and collapses to near-zero on robomimic, where the success variable is visually subdominant and the demonstrations are narrow. This both explains why the hierarchical split helps and locates the failure precisely, on a contact-rich arena (robomimic Lift/Can/Square) that the world-model planning literature barely reports.

Representation versus planning. A sharper challenge questions the planning premise itself. MR.Q [19] argues that predictive *representation* learning, not model-based planning, drives scalable multitask continuous control, exhibiting a model-free actor-critic with auxiliary predictive objectives that beats a world-model planner with no planning at all. We engage this head-on rather than against it: LeWAM’s action pathway *is* exactly such a predictive representation, and our own multi-seed results are consistent with the challenge, history-conditioned behavior cloning far outperforming online CEM on contact, while the fine A/B choices within the action head (detached versus action-grounded decoder, raw-action versus embedding-target, anti-collapse regularization) move success rate only within training-run noise. We therefore make no state-of-the-art success-rate claim. Instead we keep planning where it provably helps under the per-family rule and lean on the history-conditioned action pathway, recovering contact state, where pure search does not. The honest contribution is threefold: the per-family “when does world-model planning help” analysis; the past-action-conditioned action pathway that recovers contact state, where history-conditioned behavior cloning far exceeds CEM; and the reframing of a JEPA world model and its policy head as a single latent world-action model trained jointly, the latent-with-planning-analysis counterpart to a video-generative WAM such as DreamZero [24].

3 Method: LeWAM

Latent and dynamics. [TODO: encoder (DINOv2/ViT) $\rightarrow z$; JEPA predictor for z_{t+1} with SIGReg (isotropy $\Rightarrow$ plannable); decoder-free.]

Action pathway. [TODO: autoregressive intention head on $(z_{\leq t}, a_{< t}) \rightarrow$ action-embedding $\rightarrow$ decoder $\rightarrow$ raw action; jointly trained with the WM loss. Multitask: frozen-CLIP task token; padded+masked action dims for cross-embodiment.]

Inference modes. history-BC / intuition-guided / CEM-planning, all on one latent. [TODO: equations for the loss: $\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{sigreg}} + w_{\text{intent}} \mathcal{L}_{\text{intent}} + w_{\text{act}} \mathcal{L}_{\text{act}}$.]

4 Experiments

Setup. [TODO: LeWM latent; robomimic Lift/Can/Square (image, PH, 200 demos) + OGBench-Cube/Push-T/Reacher/Two-Room. Eval: histbc / guided / CEM-planning, $N=50$, 3 seeds; per-task budgets lift100/can240/square320.]

4.1 Planning is per-family

CEM-planning SR	goal-reaching (works)				robomimic (contact)		
	Cube	Push-T	Two-Room	Reacher	Lift	Can	Square
LeWAM (latent CEM)	76	80	90	88	0	0	0

[TODO: single-seed $N=50$ (3-seed from converging goal-reaching arms). Cube progression: 68 (LeWM repro) $\rightarrow$ 78 (planning) $\rightarrow$ 88 (intuition-guided). GC-IDM [18] reference on the same envs for context.] **Mechanism.** Both cube and robomimic are grasp tasks under an identical eval, so the contrast is not “push vs. grasp.” On cube the goal-image shows the object *displaced*, so reducing latent goal-cost requires grasping and carrying; on robomimic-lift the goal is the cube a few cm higher, visually swamped by arm pose, so CEM matches pose and skips the grasp. The failure has *two distinct causes* across the contact family. On **Lift**, the success variable (grasp of a small cube) is only weakly exposed in a single frame—a probe decodes grasp state at AUC 0.82 versus 0.99 from short history (§4.2)—so CEM matches arm pose and skips the grasp. On **Can/Square** the single-frame latent *already* decodes grasp/lift well (probe 0.96/0.999), so the cause is not perception but **multi-stage structure**: a flat horizon- H goal-image cost cannot decompose approach $\rightarrow$ grasp $\rightarrow$ transport $\rightarrow$ place, the regime hierarchical planners [7] target. Demonstrations are also narrow (200 vs $\sim$ 1M transitions). [TODO: figure: probe AUC per task (Lift 0.82 vs Can/Square 0.96/0.999); ablate aligned-goal robomimic CEM (still $\approx$ 0).]

4.2 History-conditioned BC recovers contact state

robomimic (image, PH)	Lift	Can	Square
LeWAM history-BC ($N=50$, 3-seed)	93.3	70.7	61.3
LeWAM CEM-planning	0	0	0
LDP [23] (latent diffusion planner)	69	70	46

[TODO: detach=TRUE converged ckpts; the lift/can/square 3-seed shown is EVAL-seed on one trained model (train-seed robustness per §4.3); we beat LDP on Square. Mechanism is task-dependent: on **Lift** history recovers grasp state a single frame under-exposes (probe AUC 0.82 single-frame $\rightarrow$ 0.99 latent-history $\rightarrow$ 0.999 full $[z, a]$), lifting post-grasp $P(\text{success} | \text{grasp})$ from 76 to 99; on **Can/Square** the single frame already decodes grasp (0.96/0.999), so history-BC instead supplies the sequential multi-stage execution flat CEM cannot.]

4.3 The action-head objective is within training noise

can, $N=50$	per training run	spread (all runs)
detach=TRUE (embedding-intuition)	74 / 72 / 76 = 74.0	{70.7, 80, 74, 72, 76}
solveact (raw action, $w_{\text{intent}}=0$)	70 / 70 = 70.0	{76.7, 70, 70}
detach=FALSE / anti-collapse (SIGReg)	also $\sim$ 72, within the same band	

The variants are **statistically tied** ($\sim 72 \pm 5$): the ~ 10 -point training-run variance swamps the 4–7-point effects that single runs showed. Earlier “solveact wins” / “`intent_loss` hurts” headlines were single-training-run artifacts. We additionally verify that `act_emb` does not collapse (its std *grows* to 1.28 without regularization) and that anti-collapse SIGReg does not help (80 vs. 76). We therefore report this as a negative result and caution that single-training-run A/Bs (and eval-seed-only “3-seed”) mislead.

5 Discussion

[TODO: Limitations: contribution is analysis + mechanism, not SR. Next: hierarchical subgoal-LeWAM (WM plans subgoal latents, history policy executes) to attack the contact planning= 0 regime (WorldDP/LDP-style); the two-part diagnosis motivates it precisely—subgoaling directly targets the *multi-stage* failure on Can/Square (where the latent already sees grasp), while the *perception* failure on Lift needs richer per-frame features instead. Open question: does subgoaling beat flat history-BC on the multi-stage tasks.]

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